

Evidence: plans and repeated readings

The fair-test repair lab

Name _____ Date _____

Constructed teaching cases and readings. Source and screen are 60 cm apart.

A • Flawed comparison 2 cm card: 30 cm from source. 4 cm card: 20 cm from source.	B • Repaired comparison Each card: 30 cm from source. Change height only; keep material, width and orientation the same.
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Card height	Reading 1	Reading 2	Reading 3
2 cm	4 cm	4 cm	4 cm
4 cm	8 cm	8 cm	9 cm
6 cm	12 cm	12 cm	12 cm

W1. Name the change, measurement and one control.

W2. What should the team do with the 9 cm reading?

Route 1 • Build a fair-test plan

Word bank: card height / shadow height / same / repeat.

S1. What are we deliberately changing?

- A. Card height
- B. Source-screen distance

S2. What are we measuring?

- A. Shadow height
- B. Card colour

S3. Keep each card ___ cm from the source.

S4. Do taller cards give taller or shorter shadows in the table?

S5. What should we do if a reading differs?

Route 2 • Record a method and conclusion

Use the repaired plan and readings. Optional practical: use a separate record clearly labelled “our measurements”.

T1. Write the investigation question.

Task	Plan part	Your decision
T2	Change	
T2	Measure	
T2	Keep same: two things	

T3. Write a short method including a repeat.

T4. Write a conclusion using two values from the table.

T5. Explain what to do with the 9 cm reading.

Route 3 • Review the quality of evidence

A team claims: “Our test proves every shadow is exactly twice the card height.”

X1. Use the readings to identify a problem with “exactly”.

X2. Explain why the claim cannot cover every possible setup.

X3. Write a more careful conclusion for this setup.

X4. Suggest one method check and one extra reading that would improve confidence.

Exit and reflection

Point, draw, write or tell an adult.

E1. What factor did this test change?

E2. Name one thing kept the same.

E3. Should we delete a different reading just to fit the pattern?

R1. Which idea did you improve after hearing evidence?

R2. What would you investigate or ask next?
